ISOM 672 · BUSINESS ANALYTICS · FINAL PROJECT

Who Should We Call First?

Ranking bank customers for term-deposit outreach
when contact capacity is limited

Team 10 · High Five

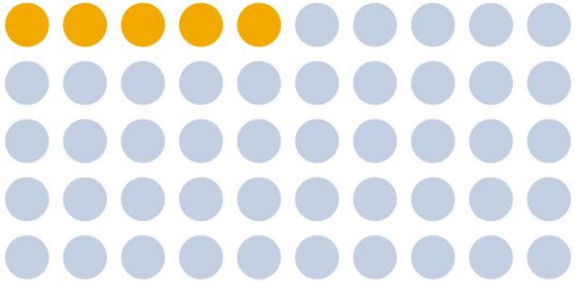
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BUSINESS UNDERSTANDING

The real decision: who goes on the call list first?

Example scenario

10,000 eligible customers, but staff time for only **1,000 calls (10%)**



Each dot ≈ 200 customers · gold = the ones we can actually call. Illustrative numbers, not dataset figures.

Not **“Will this customer subscribe?”** but **“Who do we call first?”**

1 Score

Estimate each customer's subscription score from information known before the call

2 Rank

Sort customers from highest to lowest score

3 Call from the top

Contact customers in order until capacity runs out

BUSINESS UNDERSTANDING

Objectives and how we measure success

Business objective

- More term-deposit subscriptions for the **same number of contacts**
- Less outreach spent on customers unlikely to subscribe
- Output: a ranked call list for the next campaign

Data mining objective

- Supervised binary classification: target **y** = subscribed yes / no
- Use predicted scores to **rank** customers
- Compare Logistic Regression, k-NN and Decision Tree

Precision@k

Subscribers in the top k ÷ k

k = top 10% of the list

Lift@k

Precision@k ÷ base subscription rate

How many times better than calling at random

Why not accuracy?

Predicting “no” for everyone is 88.7% accurate

…and finds zero subscribers

DATA UNDERSTANDING

UCI Bank Marketing: a Portuguese bank's phone campaigns

41,188
records · May 2008 – Nov 2010

20
predictors · 10 numeric, 10 categorical

11.27%
subscribed (4,640 records)

Target is highly imbalanced



y = subscribed to a term deposit · gold = yes, 4,640 (11.27%)

Data quality

No NaN, but an **“unknown”** category (default 20.87%, education 4.20%, housing & loan 2.40%) and **12 duplicate rows**

Four groups of predictors

Customer profile
age, job, marital, education, default, housing, loan

Current contact
contact, month, day_of_week, campaign, duration (leakage)

Campaign history
pdays, previous, poutcome

Economic context
emp.var.rate, cons.price.idx, cons.conf.idx, euribor3m, nr.employed

DATA UNDERSTANDING

What EDA showed: two traps and one real signal

Call duration = leakage

Subscribers talked 553 s vs 221 s, the biggest gap in the data. But duration is only known **after** the call.

→ **Dropped from modeling**

pdays = 999 is a code, not a number

999 means “never contacted” for 39,673 records (96.32%). Only 1,515 (3.68%) have a real gap (median 6 days in both classes).

→ **Re-coded into two features**

Real signal: history and channel

Customers whose previous campaign succeeded subscribed 65.11% of the time (vs 8.83% with no prior campaign). Cellular contacts: 14.74% vs telephone 5.23%.

→ **Kept as predictors**

DATA PREPARATION

Cleaning so the model only sees information available before the call

STEP 1
41,188 → 41,176
Removed 12 duplicate rows

→

STEP 2
Keep “unknown”
Its own category; no rows removed

→

STEP 3
Drop duration
Known only after the call (leakage)

→

STEP 4
Re-code pdays
Two clean features replace 999

pdays re-coding

pdays	previously_contacted	pdays_clean
999 (never)	0	0
6	1	6

previously_contacted = 1 if pdays ≠ 999; pdays_clean = 0 if never contacted

Model-ready data

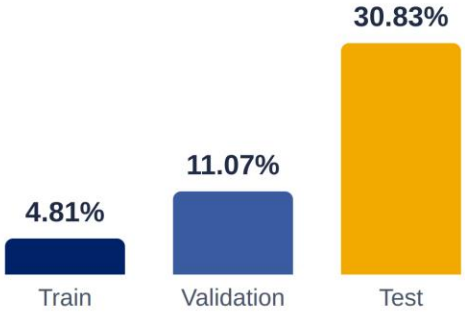
- One row = one campaign contact record; target y (yes / no)
- **41,176 records × 20 predictors**, no missing values
- One-hot encode categoricals; standardize numerics for LR and k-NN
- Every transformation fit on **training data only**

DATA PREPARATION

Time-ordered 60 / 20 / 20 split: learn from the past

Train · 60%	Validation · 20%	Test · 20%
24,705 records	8,235 records	8,236 records
May – Nov 2008	Nov 2008 – May 2009	May 2009 – Nov 2010

Subscription rate rises over time



Why not a random split?

A random split would let the model learn from future campaigns. Keeping time order mirrors real use.

What the shift means

Base rates differ by period, so we compare periods with **lift**, not raw precision. Economic indicators shifted too, so final models drop them (next slide).

MODELING

Same data, same rules, three models

Decision Tree (CART)

If-then rules from splits

k-Nearest Neighbors

Scores from similar past customers

Logistic Regression

Linear scores, regularized (L1 / L2)

15 predictors: economic indicators removed

emp.var.rate, cons.price.idx, cons.conf.idx, euribor3m and nr.employed changed sharply between 2008 and 2010. A tree using them ranked validation customers **worse than random** (AUC < 0.50).

Final models rely on customer and contact-history features, which should be more stable over time.

Model selection rule

- Pick settings by **Validation Precision@10%** (top 824 of 8,235 records)
- Ties broken by record ID (ascending)
- Imbalance: class_weight = balanced (LR, Tree); training-only oversampling (k-NN)
- Settings chosen on validation; test used for the final comparison

MODELING · DECISION TREE

Decision Tree: we fixed it, but it is still not reliable

What went wrong

Covariate shift. Economic-indicator splits learned in 2008 did not carry over: validation AUC < 0.50.

Imbalance. At 4.81% subscribers, always saying "no" is about 95% accurate and finds no one.

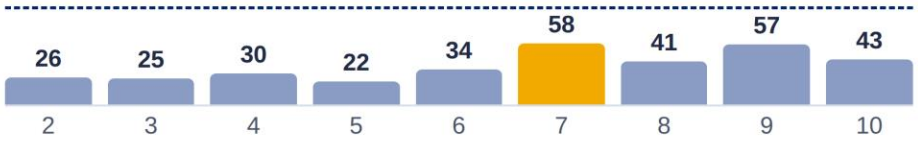
How we fixed it

- Removed the 5 economic indicators
- class_weight = balanced
- Depth 2–10, min_samples_leaf = 50, chosen by the business metric

Result

Test: subscribers in top 824	**341**
Test precision · lift	**41.38% · 1.34×**
Validation lift	**0.64×**

→ Not recommended



Fitting graph: subscribers in validation top 824

Dashed line = random (≈ 91). **No depth beats random.**

Training AUC rises 0.57 → 0.70; validation AUC stays 0.50–0.56. Deeper trees overfit.

MODELING · k-NN

k-NN: score each customer by its 25 most similar past customers

How we built it

- **Preprocessing:** standardized numerics, one-hot categoricals (distance-based model; pdays fix keeps 999 from acting like a huge gap)
- **Imbalance:** oversampled subscribers in the training set only; validation and test untouched
- **Tuning:** number of neighbors (k = 9, 15, 25), voting weights, distance metric
- **Selected:** k = 25, uniform weights, Euclidean distance

Results

Validation Precision@10%	**18.35%**
Test: subscribers in top 824	**523**
Test Precision@10% · lift	**63.47% · 2.06×**
Test ROC-AUC	**0.638**

Grid (uniform, Euclidean): k = 9 → 14.70%, k = 15 → 14.95%, k = 25 → 18.35%. At the 0.50 threshold: Recall 65.26%, F1 0.4768.

Close second on test, but a score is a vote out of 25, so only 26 values exist and ties at the cutoff can move the count (495–566).

MODELING · LOGISTIC REGRESSION

Logistic Regression: the strongest and most stable ranking

Search

20 vs 15 inputs · L1 vs L2 · none vs balanced class weight · C from 0.00001 to 1, then refined

66 candidates	**−13** all-zero models
53 eligible	**1** selected

Selected model

15 inputs · L2 · balanced · C = 0.00001

Val. subscribers, top 824	**218**
Random would find	**≈ 91**
Val. precision · lift	**26.46% · 2.39×**
Test subscribers, top 824	**550**
Test precision · lift	**66.75% · 2.17×**
Test ROC-AUC · F1@10%	**0.695 · 0.327**

Largest coefficients in the selected model

Feature	Effect on score
month = May	lower
contact = telephone	lower
month = July	higher
age (older)	lower
campaign (more calls)	lower

Ranked by coefficient size. Associations (vs. a reference level), not causes.

MODELING · COMPARISON

Three models side by side on validation

Model	Final settings	Validation Precision@10% · lift	Strengths	Weaknesses
Logistic Regression	L2, C = 0.00001, balanced; best of 53 eligible candidates	**26.46%** · 218 of 824 · **2.39×**	Interpretable coefficients; smooth scores with very few ties; cheap to retrain	Strong regularization: scores rank well but are not calibrated probabilities
k-NN	k = 25, uniform, Euclidean; scaled + one-hot; grid k = 9, 15, 25	18.35% · 151 of 823 · 1.66×	No linear assumption; uses similarity to past customers	Many tied scores; slower scoring; sensitive to scaling; hard to explain
Decision Tree	CART, depth 7, min_samples_leaf 50, balanced	7.04% · 58 of 824 · 0.64×	Easy-to-read rules	Unstable across periods; below random on validation

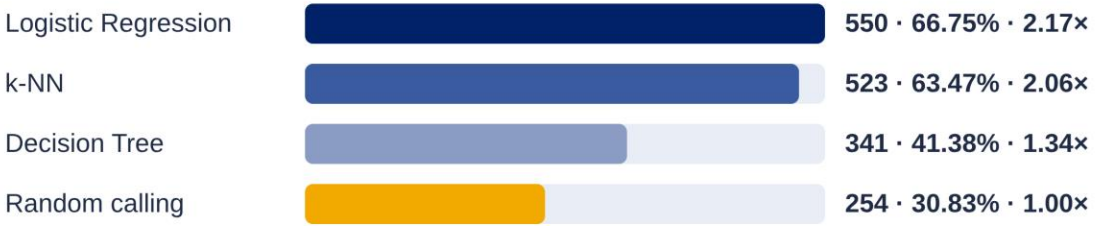
Validation base rate 11.07% (random ≈ 91 subscribers in 824). k-NN validation used the team's 823-record protocol. Tree depth 7 gave the most validation subscribers (58) even though AUC peaked at depth 4.

LR fits the business problem best: most subscribers in the top 10% of the list, few ties, readable drivers.

EVALUATION

Test period: Logistic Regression found the most subscribers

Subscribers captured in the top 824 of 8,236 test records



0.695

LR test ROC-AUC (k-NN 0.638, Tree 0.563)

0.327

LR test F1@10%

LR has the highest observed yield; k-NN remains our challenger. Ties at the cutoff can affect the comparison (k-NN: 495–566).

Historical results: they require future validation (test results were reviewed during development).

EVALUATION · BUSINESS CASE

Same 824 calls, more than twice the subscribers

+296

more historical subscribers using LR than random selection would be expected to identify (550 vs ≈ 254)

+116.5%

higher yield at equal capacity (LR vs random selection)

Business case after the pilot

ROI = (incremental subscriptions × value per subscription − added costs) ÷ added costs

Why no dollar ROI yet

- No contact costs or customer value in the data
- No implementation cost estimate
- Historical subscribers ≠ subscriptions **caused** by our call

Our alternative: a controlled pilot

- Randomly split eligible customers into two comparable pools
- Model-ranked list vs random list: same offer, capacity and outcome window
- Measure incremental subscriptions and real costs → compute ROI before scaling

DEPLOYMENT

Start with a pilot: Logistic Regression ranks, managers stay in the loop

Logistic Regression · pilot model | **k-NN · challenger** | **Decision Tree · not recommended**

1 Prepare list
Eligible customers; honor contact preferences

→

2 Score
Use only information available before outreach

→

3 Select
Highest-ranked records within capacity

→

4 Review
Manager reviews the list

→

5 Track
Contacts, costs and outcomes; monitor and decide when to retrain

Why LR: best validation result, highest test yield, clear ordering with few ties, and coefficients managers can read.

DEPLOYMENT

Risks we would manage, and ethical guardrails

Risk	Mitigation
Pre-call inputs unavailable at scoring time	Verify every input before scoring
Unseen categories (e.g. Mar, Apr, Dec absent from training)	Revalidate; retrain on data covering all months
Performance drift over time	Monitor lift at capacity; revalidate or retrain when needed

Also watched: campaign includes the final call, so its real-use effect may be weaker; tied scores at the cutoff; business impact confirmed only by the pilot.

Ethics

- Minimize data use and restrict access
- Respect opt-outs and contact limits
- Review targeting across age and job groups
- Maintain human oversight

These are proposed safeguards, not completed fairness findings.

Thank you

Questions?

Team 10 · High Five

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AI disclosure: An AI assistant (Claude) helped draft and edit the slide text and presentation. All analysis, modeling and results are the team's own work.

APPENDIX A

Logistic Regression search: best C in each group (validation)

Inputs	Penalty	Class weight	C	Precision@10%	ROC-AUC	Lift@10%
20	L1	None	1	5.10%	0.4832	0.46×
20	L1	balanced	0.001	25.97%	0.6915	2.35×
20	L2	None	0.0001	12.99%	0.5441	1.17×
20	L2	balanced	0.000001	12.99%	0.5071	1.17×
15	L1	None	0.031623	22.94%	0.6084	2.07×
15	L1	balanced	0.01	21.12%	0.6133	1.91×
15	L2	None	0.001	17.48%	0.5722	1.58×
15	**L2**	**balanced**	**0.00001**	**26.46%**	**0.6068**	**2.39×**

66 candidates (34 with 20 inputs, 32 with 15). 13 all-zero-coefficient models excluded → 53 eligible. Selected model captured 218 of 824 vs 214 for the best 20-input model; random ≈ 91.

APPENDIX B

Likely questions

Question	Short answer
Why is the test rate 30.83% but train only 4.81%?	Later campaigns simply converted more. The time-ordered split keeps that shift visible, so we compare periods with lift.
Validation lift 2.39× > test 2.17×, yet test precision is higher?	Precision depends on the base rate (11.07% vs 30.83%). Lift removes that effect.
Why keep month but drop economic indicators?	month is part of the planned outreach; the economic indicators shifted sharply and broke the tree (AUC < 0.50). Validation also favored 15 inputs (218 vs 214).
Is LR really better than k-NN?	The 27-subscriber gap is within k-NN's tie range (495–566), so LR is the pilot and k-NN the challenger.
Why show the tree at all?	It is the rejected alternative and shows why we select on validation, not test.
Are the scores probabilities?	No. C = 0.00001 plus balanced weights gives strong shrinkage; we use the scores for ranking only.

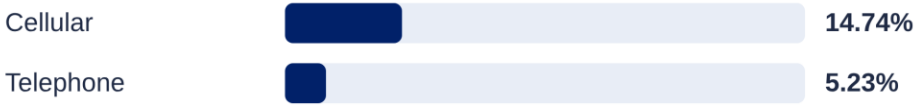
APPENDIX C · DATA UNDERSTANDING

Past success and contact channel show the clearest signals

Subscription rate by previous outcome



By contact method



Overall rate: 11.27%. Students (31.43%) and retired (25.23%) also stand out.

Numeric averages

Average	Yes	No
campaign (contacts now)	2.05	2.63
previous (past contacts)	0.49	0.13
euribor3m	2.12	3.81
nr.employed	5,095	5,176

Economic indicators differ too, but they shift over time, so final models exclude them.